Design and Implementation of a Low-Cost IoT-Based Air Purification Unit

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Abstract

In this study, a low-cost and Internet of Things (IoT)-based air purifier unit was designed and implemented. The system was created using an ESP32 microcontroller, a PM2.5 particle sensor, an MQ-135 gas sensor, a high efficiency particulate air (HEPA) filter, mini fans, and a liquid crystal display (LCD). Air quality data was collected via sensors and transmitted to a cloud platform via wireless fidelity (Wi-Fi). The fans were automatically activated when the ambient air quality exceeded the threshold values. Users were able to monitor and control the system remotely via the cloud. The developed prototype offers an effective solution to improve air quality for small businesses, clinics, and homes with its low cost and remote access features.

Keywords: IoT, Air purification unit, ESP32, PM2.5 sensor, Remote monitoring, Biomedical

1. INTRODUCTION

Air pollution is a major global concern, regardless of population density. These concerns are increasing because air pollution poses a direct threat to human health. Air pollution causes serious health problems, such as allergic reactions, nose and throat irritation, eye burning, heart disease, lung disease, asthma, and bronchitis. According to the World Health Organization (WHO), according to 2019 data, the number of people exposed to air pollution and who lost their lives was 6.7 million. In addition, indoor exposure to pollutant fuels used for cooking has reached 2.1 billion people. Approximately 90% of the world's population lives in areas where air pollution levels exceed the WHO guideline limits. More detailed information is expected to be obtained with updated data to be published in the first quarter of 2025. The current data reveals the effects of air pollution on global health [1].

The process of air purification seeks to improve air quality and safeguard human health by eliminating dangerous elements from contaminated air. Air purification systems employ a variety of methods [2, 3]. High efficiency particulate air (HEPA) filters, which can collect 99.97% of particles as fine as 0.3 microns, are a popular method. They are therefore very good at absorbing toxins and allergens [4]. Activated carbon filters are utilized in systems in addition to HEPA filters. By adsorbing pollutants and smells, activated carbon enhances air purification [5]. For air purification to be as successful as possible in a variety of settings, including hospitals, system design is essential. In this way, filtration and cleaning techniques can significantly improve air quality and health outcomes for people exposed to polluted air. When creating air filtration systems, Arduino-based systems are an affordable and adaptable option. By combining many sensors, including particulate matter (PM) and PM10/PM2.5 sensors, Arduino-based air purifiers can efficiently monitor and react to variations in air quality and create healthier indoor settings [6, 7]. The HEPA filters included in most of these air purifiers' designs are capable of removing 99.97% of particles as small as 0.3 microns. Thus, it greatly enhances the quality of the air [8]. The Arduino microcontroller's small size makes it a perfect option for creating these systems because it uses less power and operates effectively in Internet of Things (IoT) applications. Additionally, real-time pollution monitoring is made possible by the integration of air quality monitoring devices, giving customers useful information to maximize air purifier performance based on the air quality index (AQI). Fan speed and filter cleaning schedules are among the factors that these systems automatically modify to guarantee peak performance and energy efficiency. Because Arduino-based systems may be combined with cutting-edge sensor technology to produce workable indoor air quality management solutions, they are crucial. Through real-time monitoring and automated modifications, this method tackles the pressing issue of indoor air pollution, which can exacerbate health issues, and gives consumers control over their air quality [9–12]. Table 1 lists some of the Arduino work that has been completed. In the systems used within the scope of these studies, an Arduino UNO microcontroller was generally preferred. As sensors, mostly MQ series gas sensors and multi-parameter measuring sensors such as BME680 and CCS811 were used. The parameters examined included gases such as CO, CH_4, NH_3, liquefied petroleum gas (LPG), SO_2, CO_2 as well as temperature, humidity, and volatile organic compounds (VOC). In some studies, data analysis opportunities were developed with real-time monitoring and user-friendly interfaces by providing integration of mobile applications and cloud-based data access. These systems offer effective solutions, especially for the evaluation of indoor air quality and the detection of pollutant sources.

Table 1. Literature review of low-cost air quality monitoring systems

Study	Micro Controller	Sensors	Parameters Examined	Mobile Application	Notes
Precise Detection of Air Pollutants [13]	Arduino UNO	MQ-135 MQ-2 MQ-3 MQ-4 MQ-6 MQ-7	CO, CH_4, NH_3, LPG, Smoke, Alcohol, H_2S	No	Cross-sensitivity issues addressed; provides real-time ppm measurements and status messages. Designed for indoor monitoring.
Air Quality Evaluator using Arduino [14]	Arduino UNO	MQ series wireless fidelity (Wi-Fi) adapter	CO, NO, SO_2, CO_2, Temperature, Humidity	Yes	Data uploaded to the cloud; user interface available, results evaluated in Excel.
Utilizing Arduino Uno-Based System with Data Fusion [15]	Arduino UNO	BME680 CCS811	CO_2, VOC, Temperature, Humidity, Atmospheric pressure, IAQ	No	Enhanced system with data fusion and moving average filtering. Equipped with a color display and audible alarm.
Smart Indoor Air Quality Monitoring System [16]	Arduino UNO	BME680 CCS811	VOC, CO_2, Temperature, Humidity, IAQ	No	Tested in different rooms in a home environment; offers detailed evaluation based on AI-supported IAQ analysis.
Intellectual Air Quality Monitoring System Based on Arduino UNO [17]	Arduino UNO	BME680 CCS811	VOC, CO, Temperature, Humidity	No	Real-time evaluation; impact of household chemical products examined; dataset created using low-cost sensors.
IoT-Based Air Quality Monitoring System Using Arduino [18]	Arduino UNO	MQ-135 DHT11 OLED display	VOC, CO_2, Temperature, Humidity	No	Provides user-friendly interface with OLED display. Classification: Good, Poor, Very Poor, etc. Includes room ventilation suggestions.
Use of Arduino for Monitoring Indoor Air Quality [19]	Arduino UNO	Light MQ-2 Temperature Humidity	Thermal Comfort Index (ICT), Temperature, Humidity, Smoke, Light	No	Analyzed using artificial neural networks; evaluated based on thermal comfort index. Prototype developed for educational purposes.
Development of Particulate Matter Detection and Air Monitoring System by Using Arduino [12]	Arduino UNO NodeMCU ESP8266	GP2Y1010AU0F (dust sensor) MQ-2 (gas sensor - CO, LPG, smoke)	PM2.5 ($\mu g/m^3$), Smoke/gas (ppm)	Yes	Real-time notifications, low-cost prototype, data collection and comparison from different locations, environmental differences tested.
Air Quality Monitoring System in South Tangerang [20]	Arduino UNO	MQ-131(O_3) MQ-7 (CO) MQ-2 (C_4H_{10}) GP2Y1010AU0F (PM2.5)	Ozone (O_{33}), Carbon monoxide (CO), Butane (C_4H_{10}), Particulate matter (PM2.5)	No	Visual output via liquid crystal display (LCD), field-tested in urban areas, analysis-based results.
Smart Air Quality Monitoring System Using Arduino Mega [21]	Arduino Mega	Dust Sensor (probably GP2Y1010AU0F) Temperature Sensor (type not specified)	Dust density (PM), Temperature	No	LCD screen + buzzer alarm, full-day measurement, low-cost system design, alarm triggered based on temperature and particulate levels.
Development and Assessment of an Indoor Air Quality Control IoT-Based System [22]	Arduino UNO NodeMCU ESP8266	MQ-7 (CO) MQ-135 (CO_2, NH_3, NO_2, C_6H_6) DHT11 (temperature & humidity)	CO, CO_2, NO_2, NH_3, Benzene (C_6H_6), Temperature, Humidity	Yes	Energy-efficient system design, low-cost, works locally without internet connection, supports wireless control.

ESP32 microcontroller is a versatile and practical microcontroller adopted in IoT applications due to its robust connectivity, low power consumption, and cost advantages. They can be used in many areas, from healthcare systems to industrial monitoring. In this context, gas monitoring systems used in hospitals, biomedical applications that monitor physiological signals, smart home automation, and environmental sensing are microcontrollers that can be used in data processing, communication, and control processes in IoT systems [23–25]. ESP32 is also used in IoT-based air quality monitoring platforms. These systems use sensors to measure pollutants such as PM2.5, ozone, and carbon monoxide, with data transmitted to cloud servers for analysis. This setup enables real-time alerts and notifications when pollutant levels exceed safe thresholds [26]. ESP32-based systems can detect indoor air quality problems and take corrective measures, such as activating fans to increase air circulation, thus maintaining optimum indoor air quality [27]. The wireless communication capabilities of ESP32, especially the ESP-NOW protocol, enable data transmission and integration with other devices, thus enhancing the overall functionality of air purification systems [28]. In the systems used within the scope of these studies, an Arduino UNO microcontroller was generally preferred. As sensors, mostly MQ series gas sensors and multi-parameter measuring sensors such as BME680 and CCS811 were used. The parameters examined included gases such as CO, CH_4, NH_3, LPG, SO_2, CO_2 as well as temperature, humidity, and VOC. In some studies, data analysis opportunities were developed with real-time monitoring and user-friendly interfaces by providing integration of mobile applications and cloud-based data access. These systems offer effective solutions, especially for the evaluation of indoor air quality and the detection of pollutant sources.

2. MATERIAL AND METHOD

In this study, electronic and mechanical components were selected considering the criteria of low cost, easy availability, and IoT compatibility. The system design aims to fulfill both environmental data collection and active air purification functions. The selected components were determined to provide both indoor air quality measurement and environmental conditions improvement based on these data. The basic components of the proposed method are summarized in Table 2.

Table 2. Components

Component	Model/Description
Main Control Unit	ESP32 microcontroller board
PM2.5 Sensor	PM-A5 dust particle sensor
Gas Sensor (optional)	MQ-135 air quality sensor
Fans	2 mini 5V DC fans
Relay Module	2-channel 5V relay board
Filtration Element	HEPA H13 class filter
Display Unit	Nokia 5110 LCD screen
Connection Equipment	Sensor adapters, jumper wires
Power Supply	5V 2A DC adapter
Cloud Platform	Thingspeak IoT service

This system consists of three main parts: an input layer consisting of sensors that detect ambient air quality data, a control layer that processes the data, and an output layer that filters the air. This architectural structure allows for modular analysis, real-time environmental reaction, and wireless intervention—all of which contribute to a self-regulating air purification loop. The sensors (PM-A5 and MQ-135) collect data from the environment, and the ESP32 microcontroller processes these data and manages the decision mechanism. If the threshold values are exceeded, the ESP32 triggers the relay module to activate the fans and provide air flow through the HEPA filter. The measured data were also sent to the Thingspeak cloud platform. Users can monitor air quality data over the internet and manually control the fans. The general hardware architecture of the proposed system is illustrated in Figure 1. The designed and installed system is presented in Figure 2.

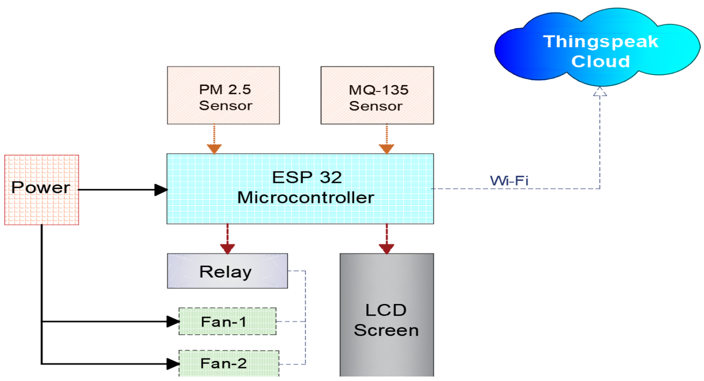

Figure 1. General system architecture of the developed IoT-based air purification unit

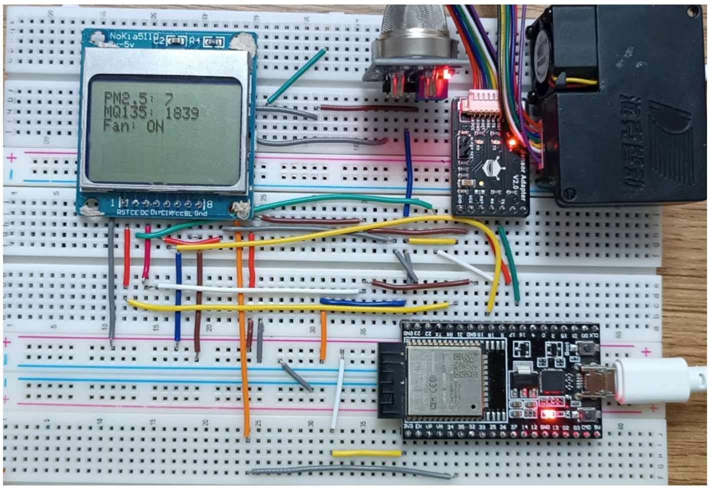

Figure 2. Developed system

In the developed system, the connections of the components to the ESP32 microcontroller board are configured as follows. The PM2.5 sensor is connected to the GPIO16 (RX) and GPIO17 (TX) pins via universal asynchronous receiver/transmitter (UART) communication. The MQ-135 gas sensor was connected to the GPIO34 pin, one of the analog-to-digital converter (ADC) inputs of ESP32, using the analog signal output. The LCD screen communicates with the SPI interface, and the CS, DC, RST, DIN, and CLK pins are assigned to GPIO5, GPIO18, GPIO23, GPIO19, and GPIO22, respectively. The relay module is controlled via two digital output pins (GPIO25 and GPIO26). The LCD screen displayed real-time PM2.5 and MQ-135 values along with the current fan activation status (ON/OFF), allowing users to interpret air quality at a glance.

When the system is turned on, the ESP32 microcontroller initializes the connected sensors, LCD display, and relay module and connects to a defined Wi-Fi network. After the initialization process is completed, the environmental data collection process begins. Air quality measurements are taken via the PM2.5 particle sensor (PM-A5) and gas sensor (MQ-135). The collected data is compared with predetermined threshold values. If the measured PM2.5 or gas level exceeds the threshold value, the relay module is triggered via the ESP32, and the fans are activated. If the measurement threshold is below the threshold, the fans remain passive. An external 5V 2A DC adapter is used to power the fans. The relay module controls the connection between the fans and the power source. Thus, the fans are powered independently from the ESP32's output, preventing any possible overcurrent issues. This setup enhances the system's stability and supports long-term operation. End of the design, developed air purification equipment shows in Figure 3.

Figure 3. Air purification device

In the system, all environmental data are transmitted to the Thingspeak cloud platform via Wi-Fi. Users can monitor instantaneous values through this platform and give remote on/off commands to fans. Data were uploaded every 15 seconds, and users could visualize real-time data and control the fans through the web interface. This interval was selected to provide sufficient temporal resolution for real-time monitoring while avoiding unnecessary network congestion and power consumption.

3. RESULTS

The IoT-based air purification system was tested in a closed simulation environment at the Afyon Kocatepe University Biomedical Laboratory. During the tests, ambient conditions were checked, sensor data were monitored in real time, and the system's behavior was evaluated. According to the data from the PM2.5 sensor, the particle density was low in the first 30 seconds of the measurement (approximately 25 µg/m³). An increase in particle density was observed starting from the 30th second, and these values gradually reached approximately 55 µg/m³ until the 60th second. With this increase, the system's fan module was automatically activated at the 60th second when the specified threshold value was exceeded. The obtained PM2.5 sensor data is presented in Figure 4.

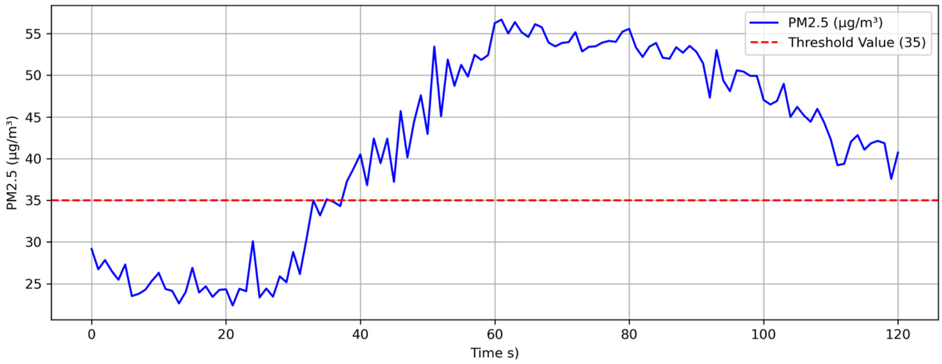

Figure 4. PM2.5 sensor data

Similarly, a significant increase was observed in the analog output values obtained from the MQ-135 gas sensor over time and stabilized at approximately 500 after the 60th second. Later, these values tended to decrease starting from the 80th second. The values of the MQ-135 gas sensor are presented in Figure 5, and the fan activation status over time is presented in Figure 6.

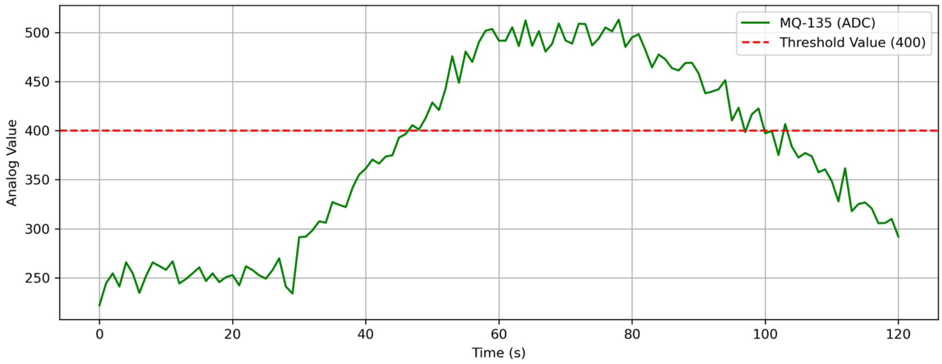

Figure 5. MQ-135 gas sensor data

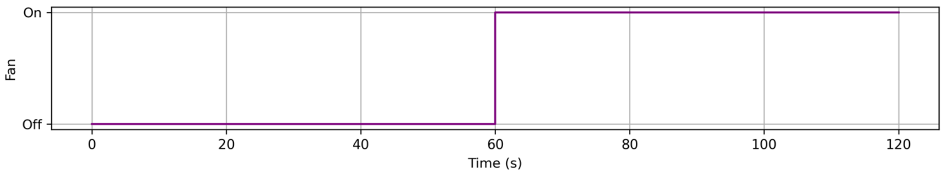

Figure 6. Fan activation status

The system sent data to the Thingspeak cloud platform every 15 seconds. The sensor data was monitored in real time via the remote monitoring screen and manual fan control was successfully provided. The system operates autonomously after initialization, continuously monitoring air quality and reacting accordingly. A view of the data collected on the ThingSpeak platform is presented in Figure 7.

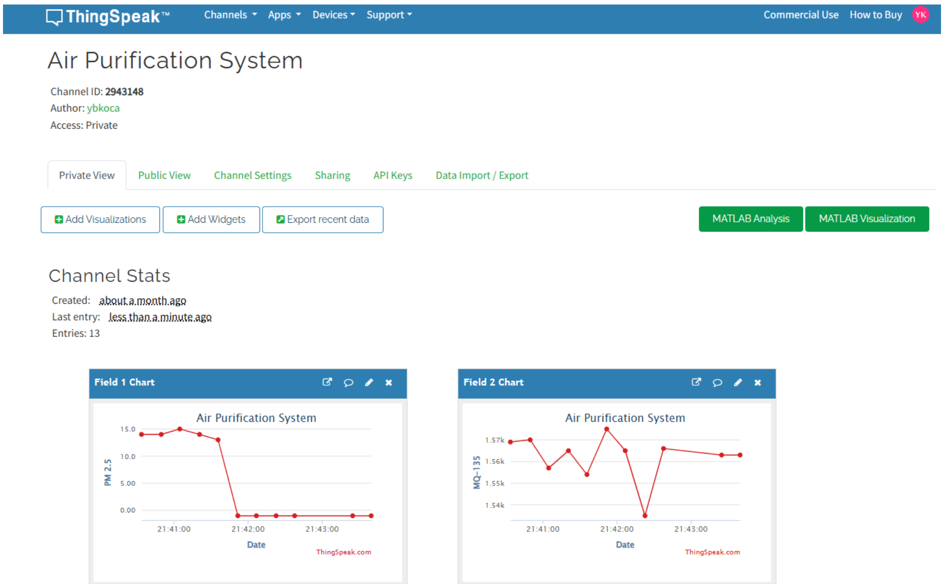

Figure 7. Air purification system data on ThingSpeak

4. CONCLUSION

In this study, a low-cost and remote-access air purification system has been designed and successfully implemented. The system detects the particle and gas density in the environment and automatically activates the fans when the threshold values are exceeded. The data obtained from the sensors is transferred to the cloud environment, and users can monitor the air quality instantly. As a result of the tests, it has been observed that the system works reliably and effectively improves the ambient air quality. It is evaluated that such systems can offer practical and economical solutions, especially in homes, small businesses, and health areas.

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